
GENERAL RELATIVITY

Tutorial problem set 7, 23.10.2020, with solutions

■ PROBLEM 19 Riemann tensor.

Riemann tensor (with all indices down) is defined to be

$$R_{\lambda\sigma\mu\nu} = g_{\lambda\rho} R^\rho_{\sigma\mu\nu} = g_{\lambda\rho} \left(\partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\alpha} \Gamma^\alpha_{\nu\sigma} - \Gamma^\rho_{\nu\alpha} \Gamma^\alpha_{\mu\sigma} \right). \quad (19.1)$$

(a) Prove that Riemann tensor is antisymmetric in its first two indices,

$$R_{\lambda\sigma\mu\nu} = -R_{\sigma\lambda\mu\nu}, \quad (19.2)$$

and that it is antisymmetric in its last two indices,

$$R_{\lambda\sigma\mu\nu} = -R_{\lambda\sigma\nu\mu}. \quad (19.3)$$

Furthermore, prove that it is symmetric under the exchange of left-hand pair and right-hand pair of indices,

$$R_{\lambda\sigma\mu\nu} = R_{\mu\nu\lambda\sigma}. \quad (19.4)$$

Finally, prove the identity

$$R_{\lambda\sigma\mu\nu} + R_{\lambda\mu\nu\sigma} + R_{\lambda\nu\sigma\mu} = 0. \quad (19.5)$$

The last identity is equivalent to writing $R_{\lambda[\sigma\mu\nu]}$. Can you explain why?

- (b) Using the symmetries of Riemann tensor derived in (a) show that Riemann tensor in D dimensions has $D^2(D^2 - 1)/12$ independent components.
- (c) In $D = 2$ dimensions Riemann tensor has only one independent component. Convince yourself that we can write it in the following form

$$R_{\lambda\sigma\mu\nu} = f(x) \left(g_{\lambda\mu} g_{\sigma\nu} - g_{\lambda\nu} g_{\sigma\mu} \right), \quad (19.6)$$

where $f(x)$ is a scalar function of coordinates. Determine what f is by requiring that the appropriate contractions of the Riemann tensor yield the Ricci scalar.

(d) Prove the Bianchi identities,

$$\nabla_\alpha R_{\rho\sigma\mu\nu} + \nabla_\mu R_{\rho\sigma\nu\alpha} + \nabla_\nu R_{\rho\sigma\alpha\mu} = 0, \quad (19.7)$$

$$\nabla^\alpha R_{\alpha\beta} - \frac{1}{2} \nabla_\beta R = 0. \quad (19.8)$$

SOLUTION: (a) We can work in a local inertial frame (LIF), in which $g_{\mu\nu} \rightarrow \eta_{\mu\nu}$, $\Gamma_{\mu\nu}^\alpha \rightarrow 0$, such that $\partial_\alpha g_{\mu\nu} \rightarrow 0$, $g^{\nu\nu} \rightarrow \eta^{\mu\nu}$ and $\partial_\alpha g^{\mu\nu} \rightarrow 0$. However, $\partial_\alpha \partial_\beta g_{\mu\nu}$ does not vanish. With these in mind, we see that the Riemann tensor in LIF can be approximated by,

$$\begin{aligned} R_{\lambda\sigma\mu\nu} &\xrightarrow{\text{LIF}} \eta_{\lambda\rho} \left\{ \partial_\mu \left[\frac{1}{2} \eta^{\rho\alpha} (\partial_\nu g_{\alpha\sigma} + \partial_\sigma g_{\nu\alpha} - \partial_\alpha g_{\nu\sigma}) \right] - \partial_\nu \left[\frac{1}{2} \eta^{\rho\alpha} (\partial_\mu g_{\alpha\sigma} + \partial_\sigma g_{\mu\alpha} - \partial_\alpha g_{\mu\sigma}) \right] \right\} \\ &= \frac{1}{2} \{ \partial_\mu \partial_\sigma g_{\nu\lambda} - \partial_\mu \partial_\lambda g_{\nu\sigma} - \partial_\nu \partial_\sigma g_{\mu\lambda} + \partial_\nu \partial_\lambda g_{\mu\sigma} \} \end{aligned} \quad (19.9)$$

From this form, it is clear that Riemann tensor in LIF possesses the symmetry, $R_{\lambda\rho\mu\nu} = -R_{\rho\lambda\mu\nu}$. To get to Riemann tensor in a general local coordinate frame, one ought perform a general coordinate transformation on (19.9). Since this transformation preserves the symmetry, we conclude that the Riemann tensor possesses the above symmetry in a general coordinate frame.

Similarly, from (19.9) one easily sees that Riemann tensor possesses also the symmetries, $R_{\lambda\sigma\mu\nu} = -R_{\lambda\sigma\nu\mu}$ and $R_{\lambda\sigma\mu\nu} = R_{\mu\nu\lambda\sigma}$ in LIF and therefore also in a general frame.

Finally, in LIF we can write Eq. (19.5) as,

$$\begin{aligned} &R_{\lambda\sigma\mu\nu} + R_{\lambda\mu\nu\sigma} + R_{\lambda\nu\sigma\mu} \\ &= \frac{1}{2} \{ \partial_\mu \partial_\sigma g_{\nu\lambda} - \partial_\mu \partial_\lambda g_{\nu\sigma} - \partial_\nu \partial_\sigma g_{\mu\lambda} + \partial_\nu \partial_\lambda g_{\mu\sigma} \} + \frac{1}{2} \{ \partial_\nu \partial_\mu g_{\sigma\lambda} - \partial_\nu \partial_\lambda g_{\sigma\mu} - \partial_\sigma \partial_\mu g_{\nu\lambda} + \partial_\sigma \partial_\lambda g_{\nu\mu} \} \\ &+ \frac{1}{2} \{ \partial_\sigma \partial_\nu g_{\mu\lambda} - \partial_\sigma \partial_\lambda g_{\mu\nu} - \partial_\mu \partial_\nu g_{\sigma\lambda} + \partial_\mu \partial_\lambda g_{\sigma\nu} \} = 0. \end{aligned} \quad (19.10)$$

By definition,

$$\begin{aligned} R_{\lambda[\sigma\mu\nu]} &= \frac{1}{6} [R_{\lambda\sigma\mu\nu} + R_{\lambda\mu\nu\sigma} + R_{\lambda\nu\sigma\mu} - R_{\lambda\sigma\nu\mu} - R_{\lambda\nu\mu\sigma} - R_{\lambda\mu\sigma\nu}] \\ &= \frac{1}{3} [R_{\lambda\sigma\mu\nu} + R_{\lambda\mu\nu\sigma} + R_{\lambda\nu\sigma\mu}]. \end{aligned} \quad (19.11)$$

The last equality follows from the antisymmetry of Riemann tensor under the exchange of the last two indices.

(b) From the first three symmetries, one concludes that the Riemann tensor $R_{\mu\nu\rho\sigma}$ in D dimensions can be considered as a tensor that is antisymmetric in the first and second pair of indices, and symmetric under the change of these two pairs of indices. The number of independent components of such a tensor is,

$$\frac{1}{2} \frac{D(D-1)}{2} \left[\frac{D(D-1)}{2} + 1 \right] = \frac{D(D-1)(D^2 - D + 2)}{8}. \quad (19.12)$$

The last symmetry (19.11), is not an independent symmetry (can you show that?). However, the smaller symmetry, $R_{[\mu\nu\rho\sigma]} = 0$ is an independent symmetry, which further restricts the number of independent components by $D \times (D-1)(D-2)(D-3)/24$. Subtracting this from (19.12) one obtains

$$\frac{D(D-1)(D^2 - D + 2)}{8} - \frac{D(D-1)(D-2)(D-3)}{24} = \frac{D^2(D^2 - 1)}{12}. \quad (19.13)$$

(c) One can easily check that the form (19.6) obeys all the symmetries discussed in part (a). Furthermore, because of metric compatibility, no derivatives acting on metric tensor can satisfy all the symmetries, making the form (19.6) unique and most general. Tracing over the first and third index gives the Ricci tensor,

$$R_{\sigma\nu} = g^{\lambda\mu} R_{\lambda\sigma\mu\nu} = f(x) g_{\sigma\nu}, \quad (19.14)$$

From here one easily obtains the Ricci scalar,

$$R = g^{\sigma\nu} R_{\sigma\nu} = 2f(x). \quad (19.15)$$

From this result we see that $f(x) = R/2$ such that the Riemann tensor in two dimensions can be also written as,

$$R_{\lambda\sigma\mu\nu} = \frac{R(x)}{2} (g_{\lambda\mu}g_{\sigma\nu} - g_{\lambda\nu}g_{\sigma\mu}). \quad (19.16)$$

(d) The Bianchi's identities (19.7) and (19.8) are easiest shown in LIR, in which the Riemann tensor is given by (19.9). In LIR the first Bianchi identity (19.7) simplifies to,

$$\begin{aligned} \nabla_\alpha R_{\rho\sigma\mu\nu} + \nabla_\mu R_{\rho\sigma\nu\alpha} + \nabla_\nu R_{\rho\sigma\alpha\mu} &\xrightarrow{\text{LIF}} \frac{1}{2} \partial_\alpha \{ \partial_\mu \partial_\sigma g_{\nu\lambda} - \partial_\mu \partial_\lambda g_{\nu\sigma} - \partial_\nu \partial_\sigma g_{\mu\lambda} + \partial_\nu \partial_\lambda g_{\mu\sigma} \} \\ &\quad + \frac{1}{2} \partial_\mu \{ \partial_\nu \partial_\sigma g_{\alpha\lambda} - \partial_\nu \partial_\lambda g_{\alpha\sigma} - \partial_\alpha \partial_\sigma g_{\nu\lambda} + \partial_\alpha \partial_\lambda g_{\nu\sigma} \} \\ &\quad + \frac{1}{2} \partial_\nu \{ \partial_\alpha \partial_\sigma g_{\mu\lambda} - \partial_\alpha \partial_\lambda g_{\mu\sigma} - \partial_\mu \partial_\sigma g_{\alpha\lambda} + \partial_\mu \partial_\lambda g_{\alpha\sigma} \} = 0. \end{aligned} \quad (19.17)$$

Upon contracting this identity with $g^{\rho\nu}g^{\rho\mu}$ one obtains,

$$\nabla_\alpha R - \nabla^\mu R_{\mu\alpha} - \nabla^\mu R_{\mu\alpha} = 0, \quad (19.18)$$

which is equivalent to (19.8).

A more general (and equally good) way of proving the first Bianchi identity is by recalling that one can define the Riemann curvature tensor in terms of the commutator of two covariant derivatives acting on a vector field. Acting covariant derivatives as in (19.7) produces a Jacobi identity, which by construction must be true. The second Bianchi identity can be subsequently proved by making a suitable contraction of the first Bianchi identity (as it is done above).

■ PROBLEM 20 Geodesic deviation.

Consider a pair of nearby freely falling particles that travel on trajectories $x^\mu(\tau)$ and $x^\mu(\tau) + \delta x^\mu(\tau)$. Their equations of motion are

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu(x) \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0, \quad (20.1)$$

$$\frac{d^2(x^\mu + \delta x^\mu)}{d\tau^2} + \Gamma_{\alpha\beta}^\mu(x + \delta x) \frac{d(x^\alpha + \delta x^\alpha)}{d\tau} \frac{d(x^\beta + \delta x^\beta)}{d\tau} = 0. \quad (20.2)$$

(a) Show that the difference between these two equations to first order in δx^μ is

$$\frac{d^2(\delta x^\mu)}{d\tau^2} + \frac{\partial \Gamma_{\alpha\beta}^\mu}{\partial x^\nu} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} \delta x^\nu + 2\Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{d(\delta x^\beta)}{d\tau} = 0. \quad (20.3)$$

(b) Show that this can be rewritten in terms of covariant derivatives along the curve $x^\mu(\tau)$,

$$\frac{D^2(\delta x^\mu)}{d\tau^2} = -R^\mu_{\alpha\nu\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} \delta x^\nu. \quad (20.4)$$

Remember the definition of the directional derivative along a curve $x^\alpha(\tau)$,

$$\frac{D}{d\tau} \equiv \frac{dx^\alpha}{d\tau} \nabla_\alpha. \quad (20.5)$$

SOLUTION: (a) Expanding Eq. (20.2) to first order in δx^μ yields,

$$\frac{d^2 x^\mu}{d\tau^2} + \frac{d^2(\delta x^\mu)}{d\tau^2} + \Gamma_{\alpha\beta}^\mu(x) \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} + \delta x^\nu \frac{\partial \Gamma_{\alpha\beta}^\mu(x)}{\partial x^\nu} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} + 2\Gamma_{\alpha\beta}^\mu(x) \frac{dx^\alpha}{d\tau} \frac{d\delta x^\beta}{d\tau} = 0. \quad (20.6)$$

Subtracting Eq. (20.1) from this equation gives (20.3).

(b) To solve this part of the problem, note first that,

$$\begin{aligned} \frac{D^2(\delta x^\mu)}{d\tau^2} &= \frac{dx^\alpha}{d\tau} \nabla_\alpha \left[\frac{dx^\beta}{d\tau} \nabla_\beta(\delta x^\mu) \right] \\ &= \frac{dx^\alpha}{d\tau} \partial_\alpha \left[\frac{dx^\beta}{d\tau} \partial_\beta(\delta x^\mu) \right] + \frac{dx^\alpha}{d\tau} \partial_\alpha \left[\Gamma_{\beta\rho}^\mu \frac{dx^\beta}{d\tau} \delta x^\rho \right] + \frac{dx^\beta}{d\tau} \Gamma_{\beta\sigma}^\mu \left[\frac{dx^\alpha}{d\tau} \partial_\alpha \delta x^\sigma + \Gamma_{\alpha\rho}^\sigma \frac{dx^\alpha}{d\tau} \delta x^\rho \right]. \end{aligned} \quad (20.7)$$

The first term is just $d^2(\delta x^\mu)/d\tau^2$ and the second can be acted the derivative, such that Eq. (20.7) can be recast as,

$$\begin{aligned} \frac{d^2(\delta x^\mu)}{d\tau^2} &= \frac{D^2(\delta x^\mu)}{d\tau^2} - (\partial_\alpha \Gamma_{\beta\rho}^\mu) \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} \delta x^\rho - \Gamma_{\sigma\rho}^\mu \frac{dx^\alpha}{d\tau} \left(\partial_\alpha \frac{dx^\sigma}{d\tau} \right) \delta x^\rho - \Gamma_{\beta\rho}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} (\partial_\alpha \delta x^\rho) \\ &\quad - \Gamma_{\beta\sigma}^\mu \frac{dx^\beta}{d\tau} \frac{dx^\alpha}{d\tau} \partial_\alpha \delta x^\sigma - \Gamma_{\beta\sigma}^\mu \Gamma_{\alpha\rho}^\sigma \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} \delta x^\rho. \end{aligned} \quad (20.8)$$

Next we make use of $\frac{d^2 x^\sigma}{d\tau^2} = -\Gamma_{\alpha\beta}^\sigma \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau}$ in the third term and note that the fourth and fifth terms nicely combine into one. Next step is to insert (20.8) into (20.3). Some terms cancel, some combine and one obtains,

$$\frac{D^2(\delta x^\mu)}{d\tau^2} + [\partial_\rho \Gamma_{\alpha\beta}^\mu - \partial_\alpha \Gamma_{\rho\beta}^\mu + \Gamma_{\rho\sigma}^\mu \Gamma_{\alpha\beta}^\sigma - \Gamma_{\alpha\sigma}^\mu \Gamma_{\beta\rho}^\sigma] \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} \delta x^\rho = 0. \quad (20.9)$$

This can be written in terms of the Riemann tensor as,

$$\frac{D^2(\delta x^\mu)}{d\tau^2} + R_{\beta\rho\alpha}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} \delta x^\rho = 0, \quad (20.10)$$

which is equivalent to Eq. (20.4). A powerful conclusion can be reached from the geodesic deviation equation, that vectors that are orthogonal to a geodesic, when parallel transported along the geodesic contain a complete information about local curvature, expressed through the Riemann tensor.

For an alternative derivation of the geodesic deviation equation, see section 3.10 of Carroll. The derivation presented here is a bit more general, since it does not rely on the assumption that δx^μ and $dx^\alpha/d\tau$ are orthogonal vectors.

■ PROBLEM 21 Weyl tensor.

The Weyl tensor in D dimensions is defined to be

$$C_{\rho\sigma\mu\nu} = R_{\rho\sigma\mu\nu} - \frac{2}{(D-2)} (g_{\rho[\mu} R_{\nu]\sigma} - g_{\sigma[\mu} R_{\nu]\rho}) + \frac{2}{(D-1)(D-2)} g_{\rho[\mu} g_{\nu]\sigma} R. \quad (21.1)$$

(a) Determine the number of independent components of the Weyl tensor in D dimensions.

Hint: Show that it has the same symmetries as the Riemann tensor, and in addition $C_{\mu\alpha\nu}^\alpha = 0$.

(b) Show that for $D \geq 3$ the Weyl tensor satisfies the following version of Bianchi identity,

$$\nabla_\rho C^\rho{}_{\sigma\mu\nu} = 2 \frac{D-3}{D-2} \left(\nabla_{[\mu} R_{\nu]\sigma} + \frac{1}{2(D-1)} g_{\sigma[\mu} \nabla_{\nu]} R \right). \quad (21.2)$$

(c) Can you argue why the expression above is not defined for $D < 3$?

SOLUTION: (a) It is easy to convince oneself that the symmetries of the Riemann tensor considered in part (a) of problem 19 are inherited by the Weyl tensor (21.1). Next, taking a trace of the Weyl tensor yields,

$$\begin{aligned} C^\mu{}_{\sigma\mu\nu} &= g^{\rho\mu} C_{\rho\sigma\mu\nu} = R_{\sigma\nu} - \frac{1}{D-2} \left((D-1) R_{\nu\sigma} - R_{\nu\sigma} + g_{\sigma\nu} R \right) + \frac{1}{(D-1)(D-2)} (D-1) g_{\nu\sigma} R \\ &= 0. \end{aligned} \quad (21.3)$$

The number of component of the Weyl tensor is then equal to the number of components of the Riemann minus that of the Ricci tensor, *i.e.*

$$\frac{D^2(D^2-1)}{12} - \frac{D(D+1)}{2} = \frac{D(D^3-7D-6)}{12} = \frac{(D+2)(D+1)D(D-3)}{12}. \quad (21.4)$$

(b) In order to arrive at the identity, notice first that from the first Bianchi identity (19.7) we have,

$$\nabla^\rho R_{\rho\sigma\mu\nu} = 2 \nabla_{[\mu} R_{\nu]\sigma}. \quad (21.5)$$

Upon making use of this and the second Bianchi identity $\nabla^\rho R_{\rho\sigma} = \frac{1}{2} \nabla_\sigma R$, one obtains

$$\nabla_\rho C^\rho{}_{\sigma\mu\nu} = 2 \nabla_{[\mu} R_{\nu]\sigma} - \frac{2}{(D-2)} \left(\nabla_{[\mu} R_{\nu]\sigma} - \frac{1}{2} g_{\sigma[\mu} \nabla_{\nu]} R \right) - \frac{2}{(D-1)(D-2)} g_{\sigma[\mu} \nabla_{\nu]} R. \quad (21.6)$$

Summing all the terms gives (21.2).

(c) From Eq. (21.4) we see that for $D = 3$ the number of components of the Weyl tensor vanishes, while when $D < 3$ it is formally negative. What this actually means is that the Weyl tensor does not exist in $D \leq 3$. The physical significance of this result is as follows. As you will learn, any the vacuum solution of the Einstein equation satisfies $R_{\mu\nu} = 0$, which also contains gravitational waves. Gravitational waves are therefore those vacuum solutions of Einstein's equations which curve space-time in such a way that they do not contribute to the Ricci tensor. Since gravitational waves curve space time, we conclude that all information about gravitational waves is stored in the Weyl tensor. Since the Weyl tensor does not exist in $D \leq 3$, that also implies that gravitational waves do not exist in $D \leq 3$. The physical explanation of that fact can be found in the spin two nature of gravitational particles, which means that they polarize space-time in such a way that the number of physical polarizations of gravitational waves in D space-time dimensions is $D(D-3)/2$, which equals to *two* in $D = 4$. Another (more pedestrian) way of seeing this arises from the observation that gravitational waves (in harmonic gauge) can be thought of as excitations of the spatial components of the metric tensor δg_{ij} satisfying $\text{Tr}[\delta g_{ij}] = 0$ and $\nabla^i \delta g_{ij} = 0$. These two conditions fix D components out of $D(D-1)/2$ components of δg_{ij} , leaving $D(D-3)/2$ components unfixed, which corresponds to the number of independent polarisations of gravitational waves. You have probably noticed that the number of independent components in gravitational waves described by metric fluctuations does not correspond to the number of independent components of the Weyl tensor. Can you think of a reason why that is the case?